吉林大学

DB2 实验报告

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姓名	柳湛博	学号	55241019
实验项目	4. Moving data		
实验性质	☐演示性实验 ☐验证性实验 ☑操作性实验 ☐综合性实验		
实验地点	A108	机器编号	10
指导教师	刘雪洁	实验时间	2026 年 9 月 26 日 12 时 25 分

一、实验综述

1. 实验目的及要求

This exercise is an online lab in which you load your tables with data using the load utility and import utility.

At the end of the lab, students should be able to:

- Use the import utility to insert data from a file into a table
- Use the load utility to fast load data from a file into a table
- Create exception tables
- Manage check constraints, triggers, and check pending status

2. 实验设备、软件

PC, windows XP Professional, DB2 9 Express-c

二、实验过程（实验步骤、记录、数据、分析）

Section 1 - Import Data in Tables

1. Import data into your ARTISTS table. For the IMPORT command, the data must be locally available to the machine from which you issue the import.

The import should specify the following:

- The import file name is artists.exp
- The import file is in ixf format
- Import in insert mode
- Write any messages to a file called art.msg

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "import from artists.exp of ixf
modified by forcein messages art.msg insert into lzb_1019.artists"

读取行数              = 79
跳过行数              = 0
插入行数              = 79
更新行数              = 0
拒绝行数              = 0
落实行数              = 79

SQL3107W  消息文件中至少有一条警告消息。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>
```

2. Look at the file art.msg to determine how many rows were inserted and any error or warning messages that may have been generated.

Show the Number of Rows Committed and messages below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type art.msg
SQL3150N  PC/IXF 文件中的 H 记录具有产品 "DB2    01.00", 日期 "19950607"
和时间 "222858"。

SQL3153N  PC/IXF 文件中的 T 记录具有名称 "art1.ixf", 限定符 "" 和源 "
"。

SQL0332N  不支持从源代码页 "850" 到目标代码页 "1386" 的字符转换。

SQL3110N  实用程序已完成处理。从输入文件读了 "0" 行。

SQL3150N  PC/IXF 文件中的 H 记录具有产品 "DB2    01.00", 日期 "19950607"
和时间 "222858"。

SQL3153N  PC/IXF 文件中的 T 记录具有名称 "art1.ixf", 限定符 "" 和源 "
"。

SQL0332N  不支持从源代码页 "850" 到目标代码页 "1386" 的字符转换。

SQL3110N  实用程序已完成处理。从输入文件读了 "0" 行。

SQL3150N  PC/IXF 文件中的 H 记录具有产品 "DB2    01.00", 日期 "19950607"
和时间 "222858"。

SQL3153N  PC/IXF 文件中的 T 记录具有名称 "art1.ixf", 限定符 "" 和源 "
"。

SQL0332N  不支持从源代码页 "850" 到目标代码页 "1386" 的字符转换。

SQL3110N  实用程序已完成处理。从输入文件读了 "0" 行。

SQL3150N  PC/IXF 文件中的 H 记录具有产品 "DB2    01.00", 日期 "19950607"
和时间 "222858"。

SQL3153N  PC/IXF 文件中的 T 记录具有名称 "art1.ixf", 限定符 "" 和源 "
"。

SQL3109N  实用程序正在开始从文件 "ARTISTS.EXP" 装入数据。

SQL3151N  因为指定了 FORCEIN 选项, 所以将不执行从 H 记录中单字节代码页值 "850"
至应用程序单字节代码页值 "1114" 的数据转换。

SQL3110N  实用程序已完成处理。从输入文件读了 "79" 行。

SQL3221W  ...开始 COMMIT WORK。输入记录计数 = "79"。

SQL3222W  ...对任何数据库更改的 COMMIT 都成功。

SQL3149N  处理了输入文件中的 "79" 行。已将 "79" 行成功插入表中。拒绝了 "0" 行。

SQL3150N  PC/IXF 文件中的 H 记录具有产品 "DB2    01.00", 日期 "19950607"
和时间 "222858"。
```

```
SQL3153N  PC/IXF 文件中的 T 记录具有名称 "art1.ixf", 限定符 "" 和源 "
"。

SQL3109N  实用程序正在开始从文件 "artists.exp" 装入数据。

SQL3151N  因为指定了 FORCEIN 选项, 所以将不执行从 H 记录中单字节代码页值 "850"
至应用程序单字节代码页值 "1114" 的数据转换。

SQL3110N  实用程序已完成处理。从输入文件读了 "79" 行。

SQL3221W  ...开始 COMMIT WORK。输入记录计数 = "79"。

SQL3222W  ...对任何数据库更改的 COMMIT 都成功。

SQL3149N  处理了输入文件中的 "79" 行。已将 "79" 行成功插入表中。拒绝了 "0" 行。


D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>_
```

3. Now you will import data into the ALBUMS and STOCK tables using script files. The file you will use to import data into the ALBUMS table is called imp_albu.

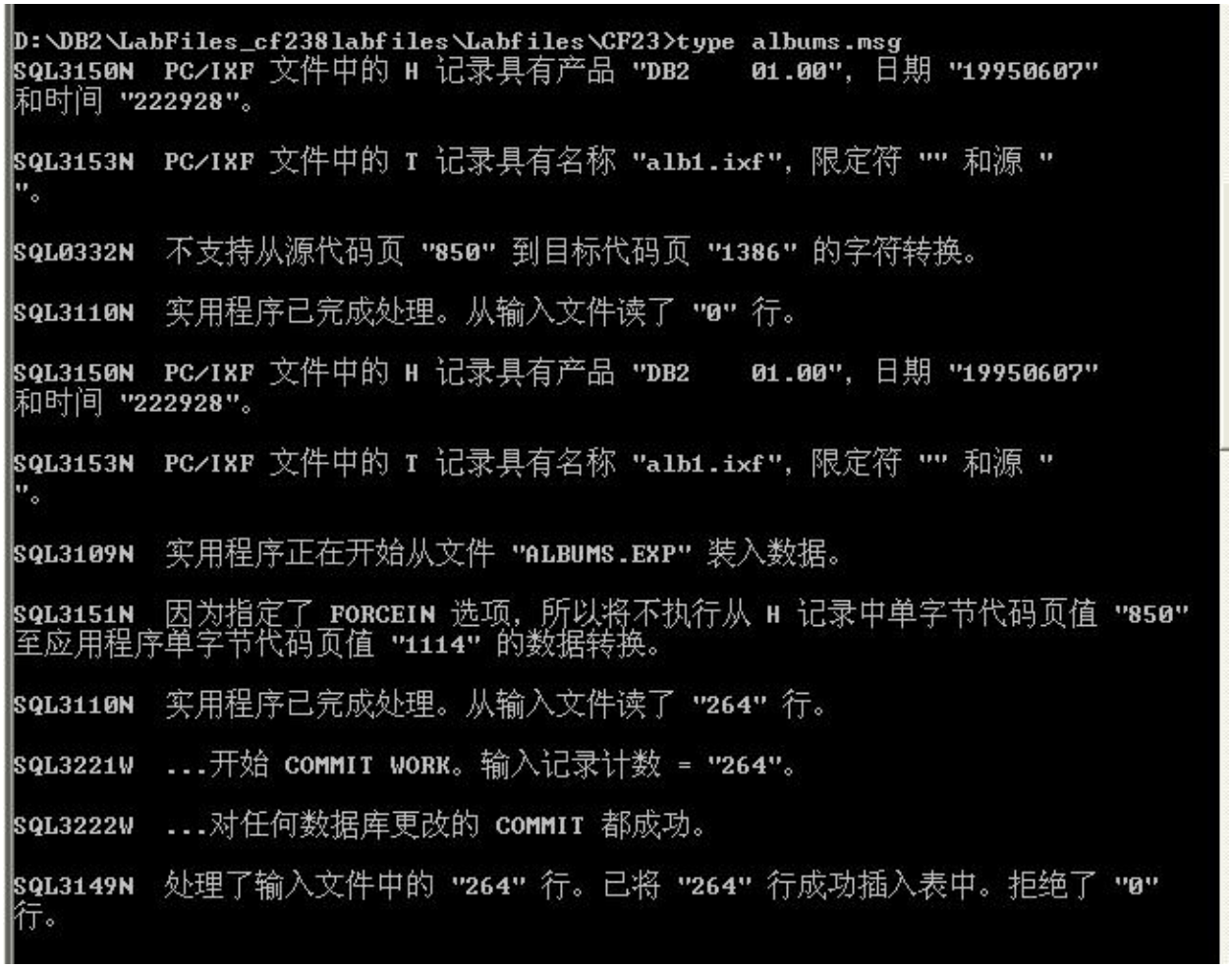
Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf imp_albu
import from D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23\albums.exp of ixf modifi
ed by forcein messages albums.msg insert into albums

读取行数          = 264
跳过行数          = 0
插入行数          = 0
更新行数          = 0
拒绝行数          = 264
落实行数          = 264

SQL3107W  消息文件中至少有一条警告消息。
```

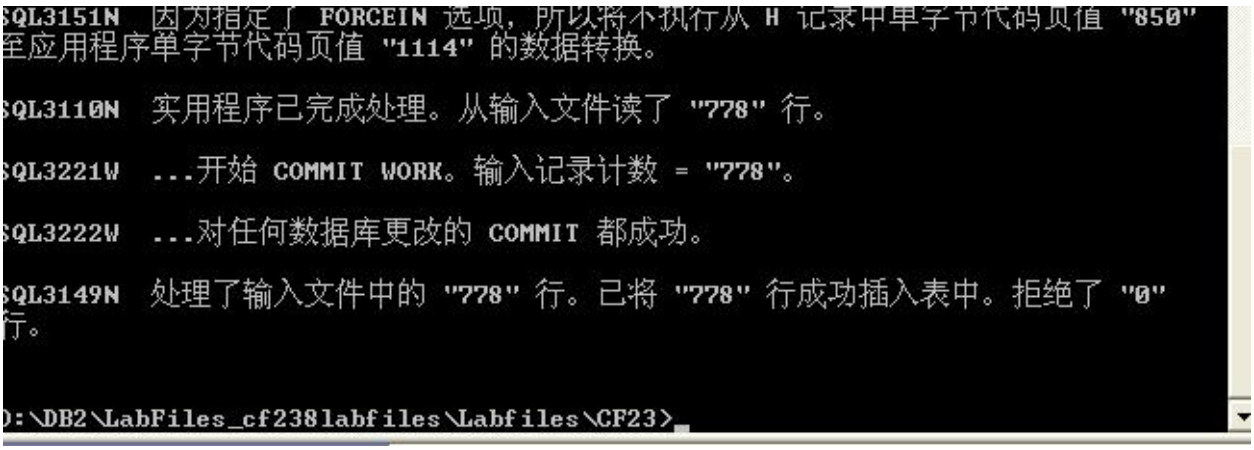
4. Record the number of rows committed. Check albums.msg to see if any additional messages occurred.
Show your work below by printing screen.



5. The file you will use to import data into the STOCK table is called imp_sto.
Show your work below by printing screen.



6. Record the number of rows committed. Check sto.msg to see if any additional messages occurred.
Show your work below by printing screen.



7. For all the imports, where was the source file located that was being imported, on the Windows machine or on the UNIX Database Server?
IMPORT 由当前 Windows DB2 Command Window 执行，源文件位于 Windows 客户端机器，而不是 UNIX Database Server。IMPORT 从客户端读取源文件后导入数据库。

8. Remember that at the end of the previous lab, you created a trigger on the STOCK table that should put rows into the REORDER table if any rows were updated to have a quantity of less than or equal to 5 in the STOCK table.
Check to see if the trigger was fired during the Import utility.
Show your work below by printing screen.
Explain what is returned.
本次 IMPORT 使用 INSERT 模式，不会触发只针对 QTY 更新定义的 REORDER 触发器，因此不会产生由本次 IMPORT 新增的补货记录。



Section 2 - Create Exception Tables - ARTISTS, ALBUMS, STOCK

1. We have provided a script named crexptab that will create exception tables for ARTISTS, ALBUMS, and STOCK. Make sure each table is created successfully.

Show your work below by printing screen

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf CREXPTAB
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


create regular tablespace sms02 managed by system using ('D:\sms\sms04', 'D:\sms
\sms05', 'D:\sms\sms06')
DB20000I  SQL 命令成功完成。

create table artexp (artno          smallint not null, name           varchar (5
0), classification char (1) not null, bio            clob (100K) logged compact,
 picture        blob (500k) not logged compact, ts             timestamp, msg
         clob (32K)) in sms02
DB20000I  SQL 命令成功完成。

create table albexp (title     varchar (50), artno     smallint not null, itemno
    smallint not null, ts        timestamp, msg        clob (32K)) in sms02
DB20000I  SQL 命令成功完成。

create table stoexp (itemno             smallint not null, type               ch
ar (1) not null, price             decimal (5,2) not null with default, qty
           int not null with default, ts                 timestamp, msg
        clob (32K)) in sms02
DB20000I  SQL 命令成功完成。
```

Note that the table space SMSEXP is being created to store all of the exception tables. Also note that the exception tables mimic the base table plus two additional columns.

What are the exception table names? What are the two additional columns and their attributes? Are they optional?
ARTEXP
ALBEXP
STOEXP

列名	类型	作用
TS	TIMESTAMP	异常行进入异常表的时间
MSG	CLOB(32K)	约束违规或装载失败原因

TS、MSG 不是基础业务表字段，但在本实验的异常表中用于保存时间和错误信息。重新插入原表时不能插入这两列。

Section 3 - Backup Database

1. The backup will be done against your database server and will use a directory path of C:\backup. Using your Command Window, enter the following commands:

```
cd mkdir backup
db2 force application all
db2 "backup db musicdb to C:\backup"
```

The backup will take several minutes to run.

Show your work below by printing screen.



2. Verify the backup completed successfully by checking the messages that appear on your screen. What is its status?

Show your work below by printing screen.

```
D:\>db2 list history backup all for lzb_1019

                    列示 lzb_1019 的历史记录文件

匹配的文件条目数 = 1

 Op 对象 时间戳记+序列      类型 设备 最早日志     当前日志     备份标识
 -- --- ------------------ ---- --- ------------ ------------ --------------
  B  D  20261011092935001   F    D  S0000000.LOG S0000000.LOG
 ----------------------------------------------------------------------------
  包含 10 表空间:

  00001 SYSCATSPACE
  00002 USERSPACE1
  00003 DMS01
  00004 DMS02
  00005 DMS03
  00006 DMS04
  00007 DMS05
  00008 DMS06
  00009 SMS01
  00010 SMS02
 ----------------------------------------------------------------------------
    注释: DB2 BACKUP LZB_1019 OFFLINE
 开始时间: 20261011092935
   结束时间: 20261011092938
     状态: A
 ----------------------------------------------------------------------------
  EID: 12 位置: D:\backup


D:\>dir D:\backup
 驱动器 D 中的卷没有标签。
 卷的序列号是 000B-F297

 D:\backup 的目录

2026-10-11  09:29    <DIR>          .
2026-10-11  09:29    <DIR>          ..
2026-10-11  09:29       107,073,536 LZB_1019.0.DB2.NODE0000.CATN0000.20261011092
935.001
               1 个文件    107,073,536 字节
               2 个目录 19,112,558,592 可用字节
```

Section 4 - Load Data with INSERT option into CONCERTS

1. LOAD data into your CONCERTS table.

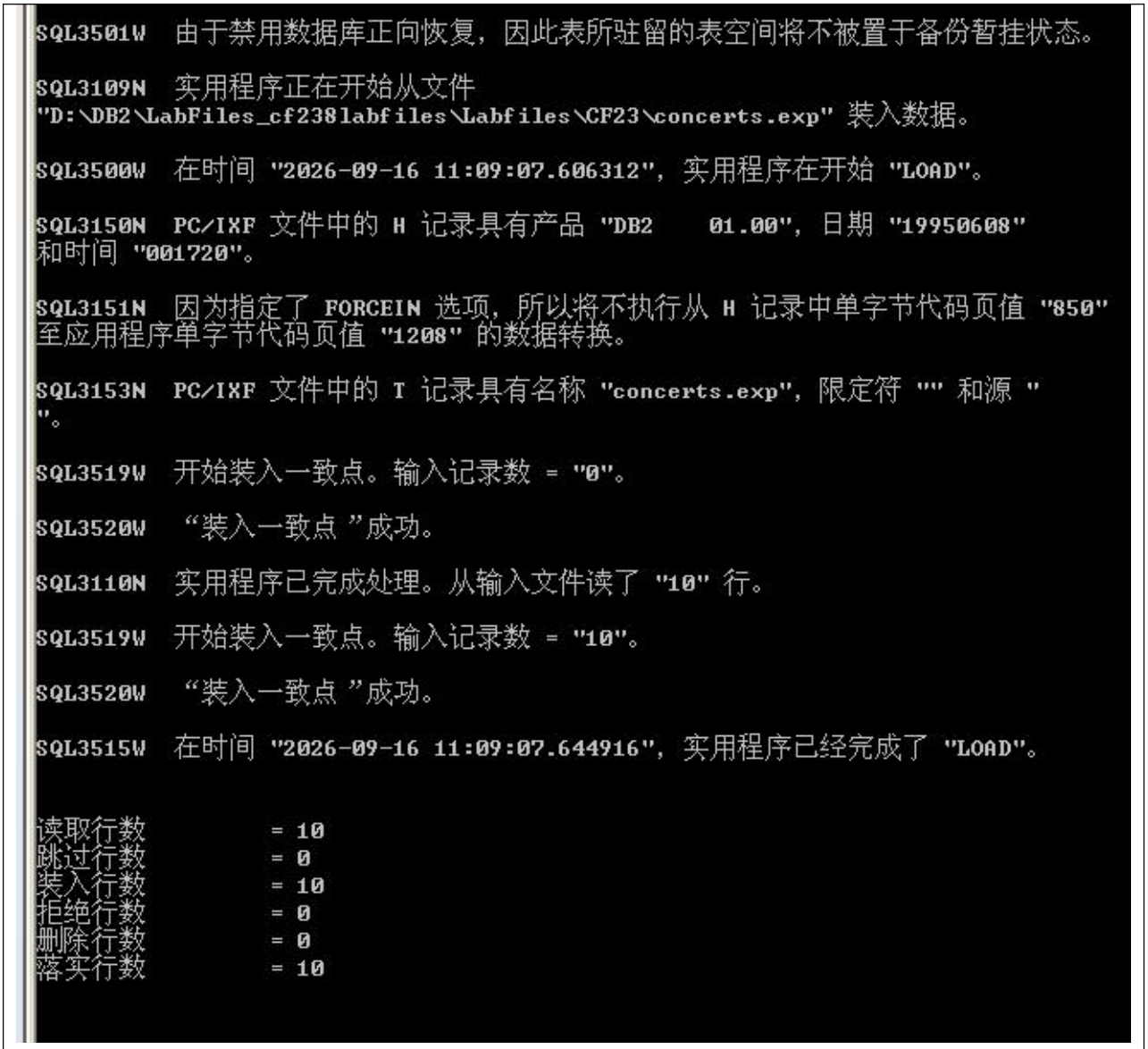
The load should specify the following:

- •Work with files in the path C:\labfiles\cf23
- •The load file name is concerts.exp
- •The import file is in ixf format
- •Load in insert mode
- •Write any messages to a file called concerts.msg

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "load from concerts.exp of ixf m
odified by forcein messages concerts.msg insert into lzb_1019.concerts"

读取行数          = 10
跳过行数          = 0
装入行数          = 10
拒绝行数          = 0
删除行数          = 0
落实行数          = 10
```



2. Record the number of rows successfully loaded and committed from the messages file.

Show your work below by printing screen.

```
读取行数          = 10
跳过行数          = 0
装入行数          = 10
拒绝行数          = 0
删除行数          = 0
落实行数          = 10
```

Section 5 - Load Data with REPLACE option into ARTISTS

1. LOAD data into the ARTISTS table using a script file named load_art.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf LOAD_ART
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


load from art2.exp of ixf modified by forcein messages load_art.msg replace into
 artists for exception artexp

读取行数          = 78
跳过行数          = 0
装入行数          = 78
拒绝行数          = 0
删除行数          = 0
落实行数          = 78
```

2. The LOAD command loads rows into the ARTISTS table using a different IXF

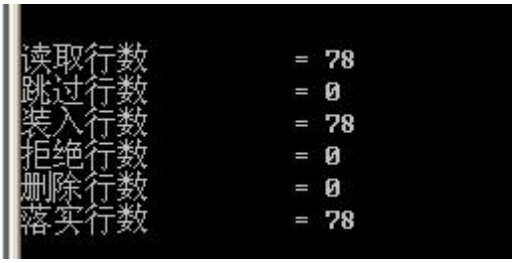
input file from the earlier IMPORT.

Note that it uses the REPLACE option to replace the previously imported rows in the ARTISTS table. The artexp table is specified to contain exception rows for the ARTISTS table during the load with the FOR EXCEPTION OPTION.

What type of violations will be stored in this table during the load?

在 LOAD 执行期间，`ARTEXP` 例外表中将保存违反唯一性约束（Unique Key Violations）或主键冲突（Primary Key Violations）的重复记录。原理剖析：LOAD 实用程序在执行的第 3 阶段（Delete 阶段）会基于 Build 阶段生成的索引检测唯一键冲突。若数据中包含主键 `ARTNO` 重复的行，冲突行将被移出目标表并写入 `ARTEXP` 中，同时在 `TS` 记录发生时间，在 `MSG` 记录违规原因代码。在 LOAD 阶段不会检查外键约束，外键违规不会进入此时的例外表。

3. Record the number of rows committed.



4. Check the ARTEXP table to see if any exception rows violated the unique key index on the ARTISTS table. Use the script file selexp.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf SELEXP
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


select artno, ts, substr(msg,1,40) from artexp

ARTNO  TS                          3
------ --------------------------- ----------------------------------------

  0 条记录已选择。


select artno, itemno, ts, substr(msg,1,40) from albexp

ARTNO  ITEMNO TS                             4

------ ------ --------------------------- ----------------------------------------
-

  0 条记录已选择。


select itemno,ts, substr(msg,1,40) from stoexp

ITEMNO TS                          3
------ --------------------------- ----------------------------------------

  0 条记录已选择。
```

查询返回 `0 条记录已选择`。这表明输入的 `art2.exp` 文件中的 78 行数据彼此之间不存在重复的主键 `ARTNO`，没有发生任何违反唯一键索引的冲突，因此 `ARTEXP` 例外表当前为空。

5. Check the load_art.msg file for messages.
Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type load_art.msg
SQL3501W  由于禁用数据库正向恢复，因此表所驻留的表空间将不被置于备份暂挂状态。

SQL3109N  实用程序正在开始从文件
"D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23\ART2.EXP" 装入数据。

SQL3500W  在时间 "2026-09-16 00:10:20.626983"，实用程序在开始 "LOAD"。

SQL3150N  PC/IXF 文件中的 H 记录具有产品 "DB2    01.00"，日期 "19950607"
和时间 "230719"。

SQL3050W  将在 IXF 文件代码页 "850" 与应用程序代码页 "1208" 之间进行转换。

SQL3153N  PC/IXF 文件中的 T 记录具有名称 "art2.exp"，限定符 "" 和源 "
"。

SQL3519W  开始装入一致点。输入记录数 = "0"。

SQL3520W  “装入一致点 ”成功。

SQL3110N  实用程序已完成处理。从输入文件读了 "78" 行。

SQL3519W  开始装入一致点。输入记录数 = "78"。

SQL3520W  “装入一致点 ”成功。

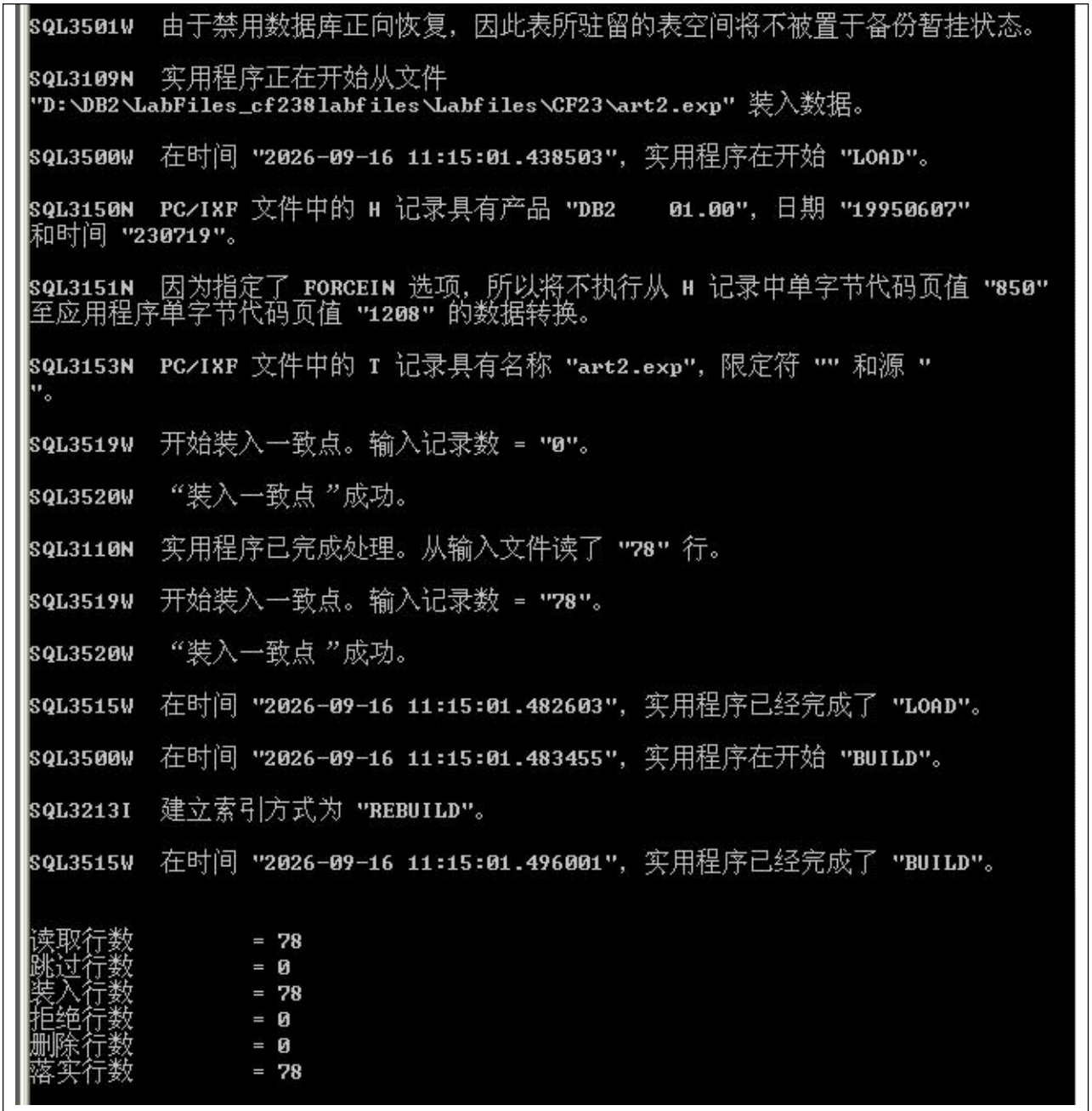
SQL3515W  在时间 "2026-09-16 00:10:20.659571"，实用程序已经完成了 "LOAD"。

SQL3500W  在时间 "2026-09-16 00:10:20.660127"，实用程序在开始 "BUILD"。

SQL3213I  建立索引方式为 "REBUILD"。

SQL3515W  在时间 "2026-09-16 00:10:20.673639"，实用程序已经完成了 "BUILD"。


读取行数          = 78
跳过行数          = 0
装入行数          = 78
拒绝行数          = 0
删除行数          = 0
落实行数          = 78
```



Section 6 - Examining the SET INTEGRITY PENDING State

1. Try to select from the ARTISTS, STOCK, and ALBUMS tables using the script seltab.

Show your work below by printing screen.



2. What SQL code did you get on the first select statement? Check the meaning of the SQL code using the online help facilities. What action do you need to take to clear this status?

Answer question and show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "? SQL0668N"


SQL0668N  不允许对表 "<表名>" 执行操作，原因码为 "<原因码>"。

说明:

限制对表 "<表名>" 的访问。原因基于下列原因码 "<原因码>":

1

        该表处于"设置完整性暂挂无访问"状态。未强制表的完整性并且表的内
        容可能是无效的。如果从属表处于"设置完整性暂挂无访问"状态，那么
        对于未处于"设置完整性暂挂无访问"状态的父表或基础表执行的操作也
        可能会接收到此错误。


2

        表处于"无数据移动"状态。当处于此状态时，禁止导致数据移动的操作
        。数据移动操作包括 REDISTRIBUTE、数据库分区键的更新、多维集群
        键的更新、范围集群键的更新、数据分区键的更新和 REORG TABLE。


3

        表处于"装入暂挂"状态。先前尝试装入（LOAD）此表失败。在重新启动
        或终止 LOAD 操作之前不允许对表进行访问。


4

        表处于"读访问"状态。此状态可以在联机装入（LOAD）处理（带有 READ
        ACCESS 选项的 LOAD INSERT）期间发生，或在联机装入（LOAD）操作
        后发生，除了在使用 SET INTEGRITY 语句在表的新追加的部分验证所
        有约束之前。不允许对此表的更新活动。


5

        表处于"正在装入"状态。LOAD 实用程序当前对此表进行操作，直到
        LOAD 完成才允许访问。


6

        不能在 ESE 中刷新引用昵称的具体化查询表。
```

7

表处于"REORG 暂挂"状态。在执行包含 REORG 建议的操作的 ALTER TABLE 语句后，可能会发生这种情况。

8

表处于"改变暂挂"状态。当在包含 REORG 建议的操作的 ALTER TABLE 语句所在工作单元中使用该表时，就可能会发生这种情况。

9

表处于"重新分发暂挂"状态。REDISTRIBUTE 实用程序未完成对此表的操作，直到 REDISTRIBUTE 完成才允许访问。

10

该表是正在执行的 ADMIN_MOVE_TABLE 操作的源表。所尝试的操作在移动操作完成或被取消前将受限制。

用户响应：

1

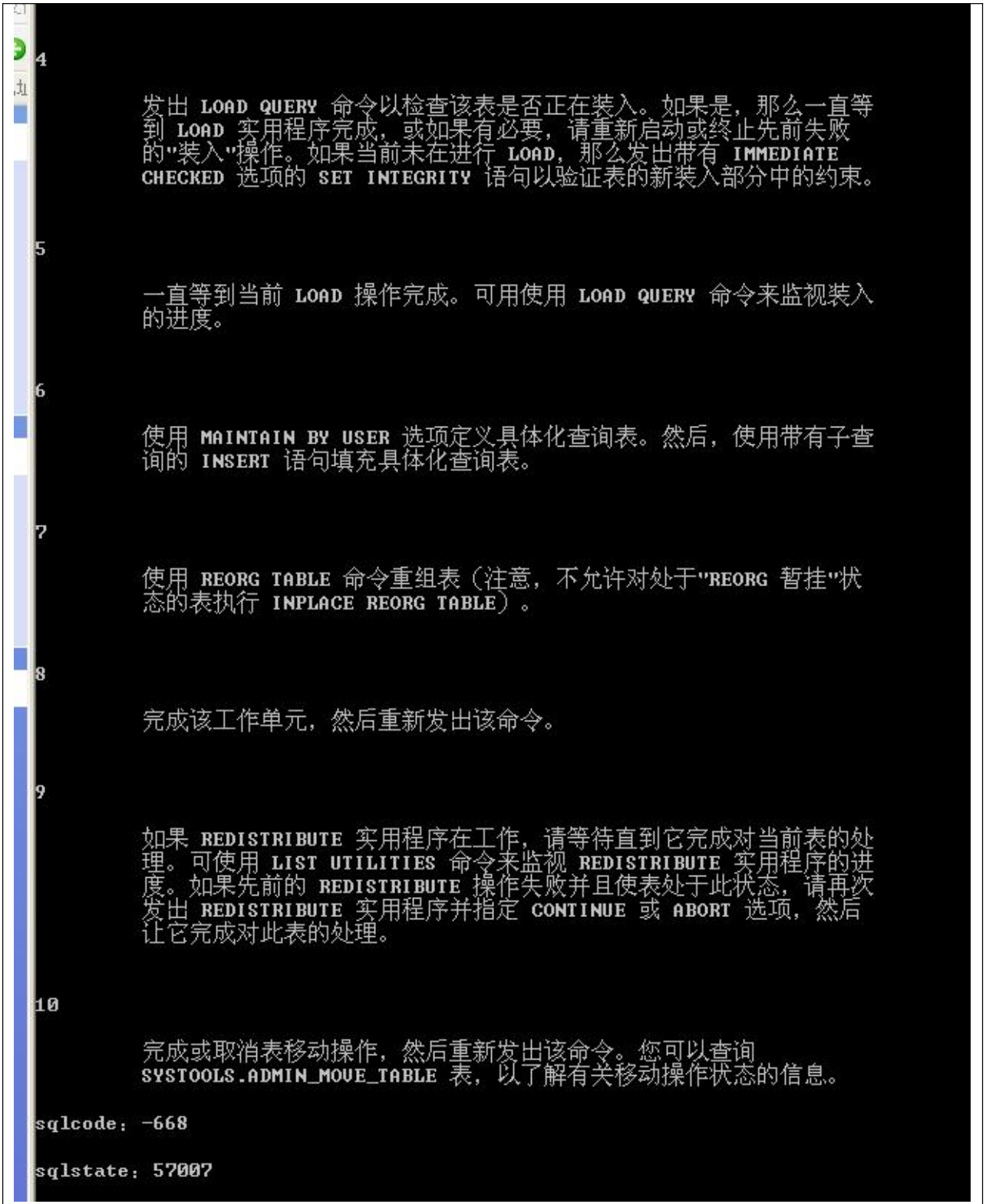
对表 "<表名>" 执行带有 IMMEDIATE CHECKED 选项的 SET INTEGRITY 语句，以使表脱离"设置完整性暂挂无访问"状态。对于用户维护的具体化查询表，执行带有 IMMEDIATE UNCHECKED 选项的语句，而不是带 IMMEDIATE CHECKED 选项。

2

对表 "<表名>" 的从属立即具体化查询表和登台表执行 REFRESH TABLE 语句。可以通过先前的 LOAD INSERT 操作根据 "<表名>" 的追加数据以及通过先前带有 ATTACH 子句的 ALTER TABLE 语句根据 "<表名>" 的连接数据以增量方式维护这些从属立即具体化查询表和登台表的内容。

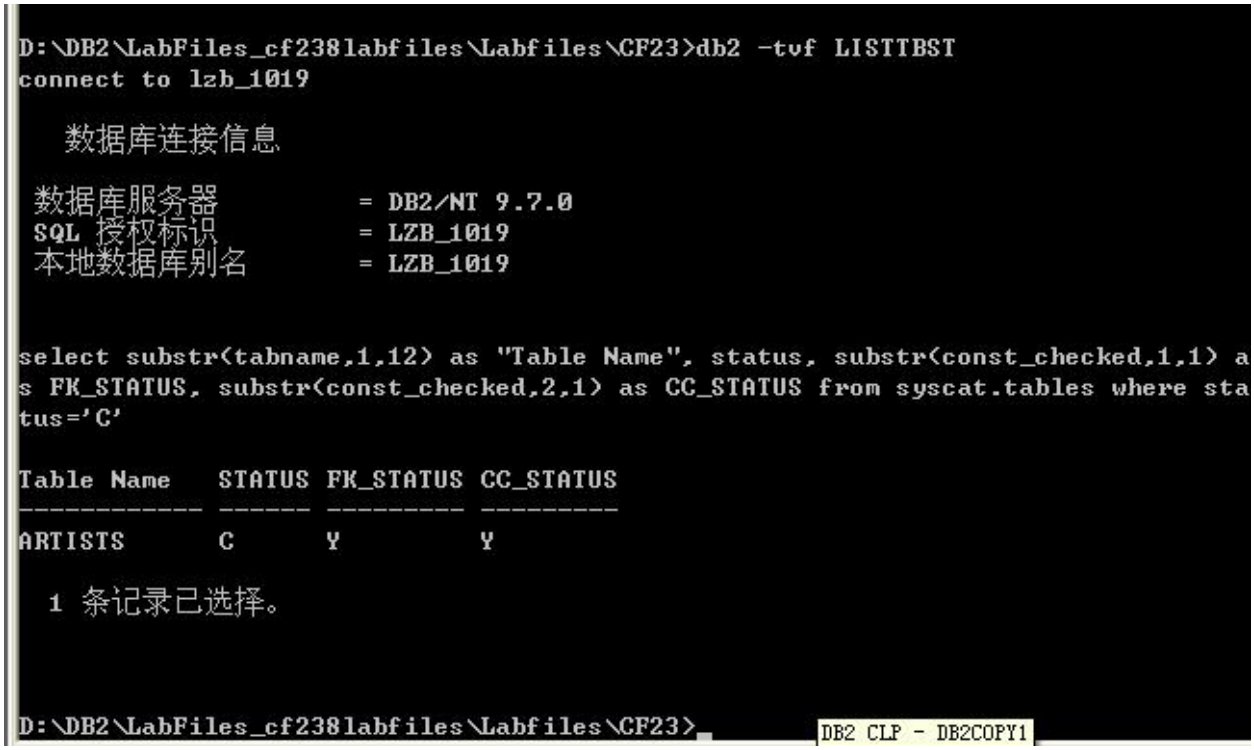
3

通过分别发出带有 RESTART 或 TERMINATER 选项的 LOAD 来重新启动或终止先前失败的对此表的 LOAD 操作。



3. Check the check constraints status of the tables by executing the script listtbst and record the status.

Show your work below by printing screen.



4. The SET INTEGRITY SQL statement can be used to check for referential constraint exceptions. Exception tables should be specified for all tables in a check pending state (in this case, just ARTISTS).

Run the script setcsts_art to check for referential constraint exceptions in the ARTISTS table.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf setcsts_art
connect to lzb_1019

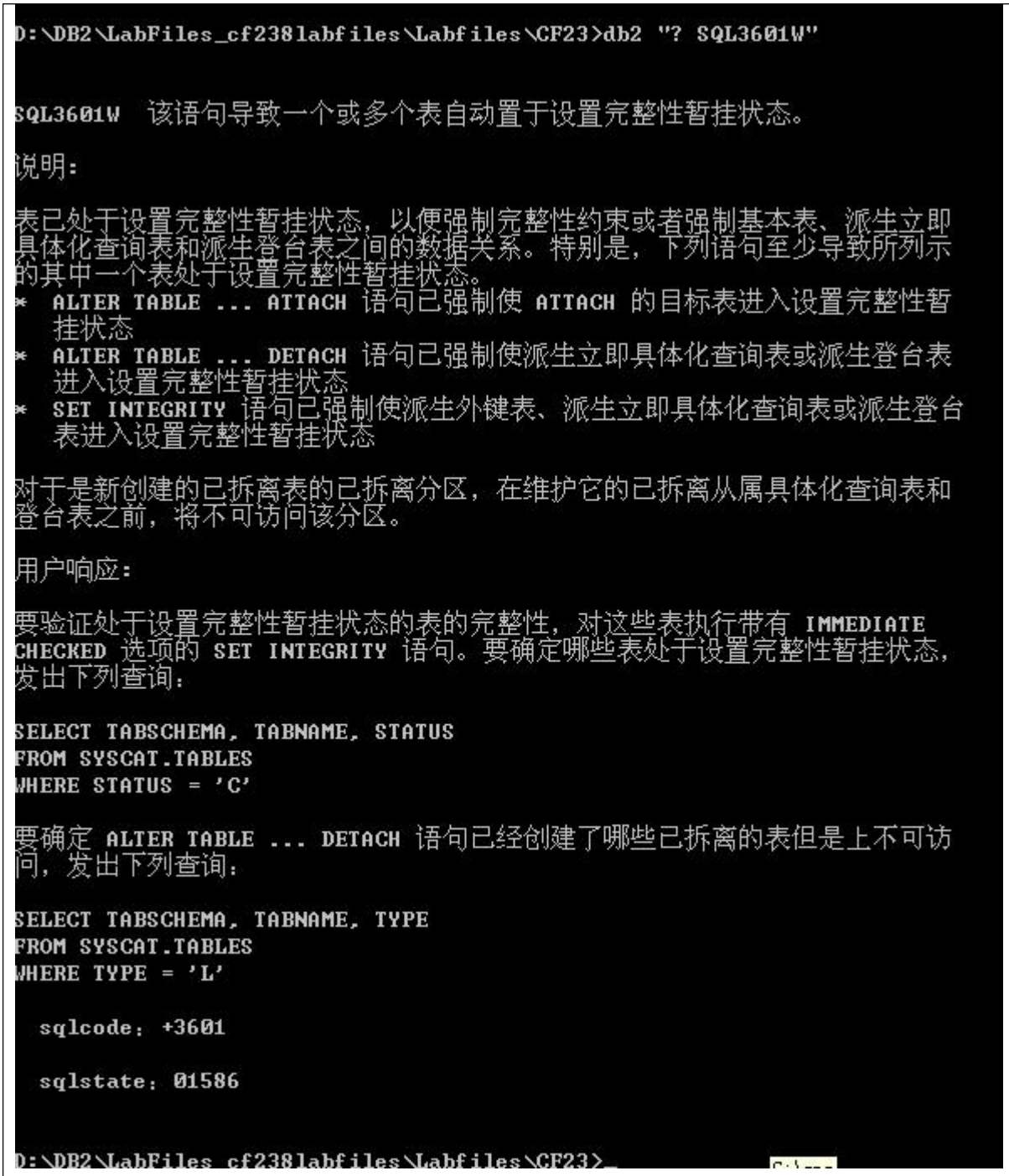
   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


set integrity for artists immediate checked for exception in artists use artexp
SQL3601W  该语句导致一个或多个表自动置于设置完整性暂挂状态。  SQLSTATE=01586
```

5. What does the warning SQL3601 mean?

Show your work below by printing screen.



6. Check the check constraints status of the tables by executing the script listtbst and record the status.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf LISTTBST
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


select substr(tabname,1,12) as "Table Name", status, substr(const_checked,1,1) a
s FK_STATUS, substr(const_checked,2,1) as CC_STATUS from syscat.tables where sta
tus='C'

Table Name   STATUS FK_STATUS CC_STATUS
------------ ------ --------- ---------
ALBUMS       C      N         Y
STOCK        C      N         Y

  2 条记录已选择。
```

* 父表 `ARTISTS` 已经不在列表中（`STATUS` 恢复为 `'N'` 正常状态）。
* 子表 `ALBUMS`和 `STOCK`现在处于 `STATUS = 'C'`（Check Pending），且 `FK_STATUS = 'N'`（外键尚未校验通过）。

7. The SET INTEGRITY SQL statement can be used to check for referential constraint exceptions. Exception tables should be specified for all tables in a check pending state (in this case, ALBUMS and STOCK). Run the script setcsts_2 to check for referential constraint exceptions in the ALBUMS and STOCK tables.
Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf setcsts_2
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


set integrity for albums, stock immediate checked for exception in albums use al
bexp, in stock use stoexp
SQL3602W  检查数据处理找到约束违例，并将它们移至异常表。  SQLSTATE=01603
```

在执行该语句时，DB2 逐行核对 `ALBUMS` 和 `STOCK` 的外键。DB2 发现：

1. `ALBUMS` 中有一张专辑（关联 `ARTNO = 100`），在父表 `ARTISTS` 中

找不到匹配的父键！

2. `STOCK` 中有一条库存（关联 `ITEMNO = 300`），同样找不到对应的父键！

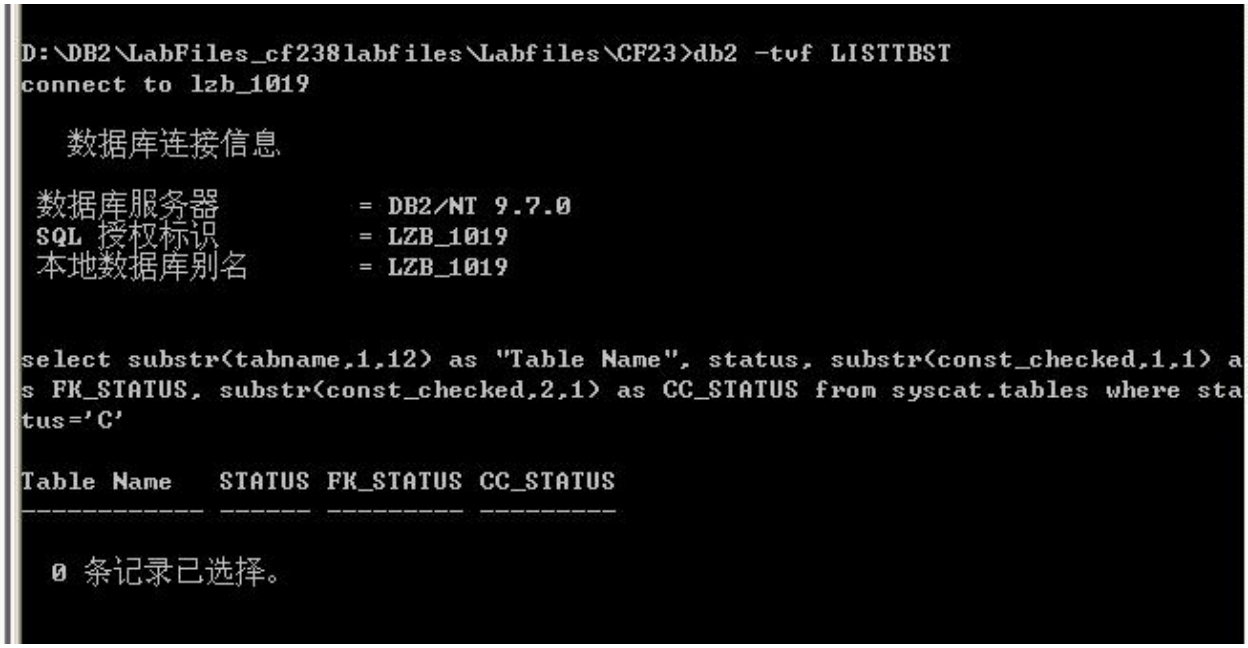
3. DB2 并没有直接报错中止，而是严格遵循 `FOR EXCEPTION` 指示，自动将这两条违反外键约束的孤立行从基础表中移出，安全投递到了例外表 `ALBEXP` 和 `STOEXP` 中！

4. 基础表中剩下的全部是合规行，暂挂状态因此得以顺利清除！

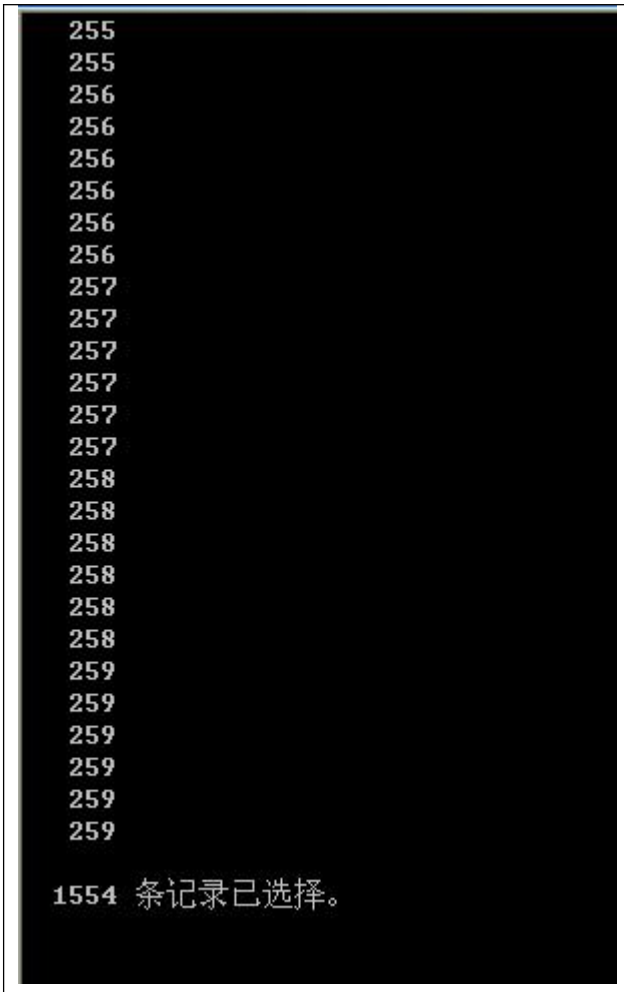
8. Select from the tables to verify that the check pending status has been removed from the tables. Run the scripts listtbst and seltab.

Were there any constraint pending conditions?

Answer question and show your work below by printing screen.







没有剩余的约束暂挂状态（No constraint pending conditions remained）。
执行 LISTTBST 查询返回 0 条记录已选择，证明 ARTISTS、ALBUMS 和 STOCK 三张表的 STATUS 均已从 'C' 恢复为正常的 'N'。随后对各表执行 SELECT 查询均能正常返回数据，不再出现 SQL0668N 错误，完整性暂挂状态已彻底解除。

9. Where are the rows that were moved out of the tables in order to get the table out of constraint pending? How can you see them?
Show your work below by printing screen.

违规行去向：被移出的行被 DB2 自动转存到了对应的例外表（Exception Tables）中：

从 `ALBUMS` 表移出的行保存在 `ALBEXP`中；

从 `STOCK` 表移出的行保存在 `STOEXP` 中。

如何查看：可以通过直接对这两个例外表执行标准的 SQL `SELECT` 查询来查看违规数据，例如运行脚本 `db2 -tvf SELEXP`，或者执行：

```
SELECT * FROM LZB_1019.ALBEXP;
SELECT * FROM LZB_1019.STOEXP;
```

10. Examine the rows that were moved to your exception tables using the script file selexp.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf SELEXP
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


select artno, ts, substr(msg,1,40) from artexp

ARTNO  TS                          3
------ --------------------------- ----------------------------------------

  0 条记录已选择。


select artno, itemno, ts, substr(msg,1,40) from albexp

ARTNO  ITEMNO TS                          4

------ ------ --------------------------- ----------------------------------------
-
   100    300 2026-10-11-15.34.02.305000 00001F00023LZB_1019.ALBUMS.FKARTNO

   100    301 2026-10-11-15.34.02.305000 00001F00023LZB_1019.ALBUMS.FKARTNO


  2 条记录已选择。


select itemno,ts, substr(msg,1,40) from stoexp

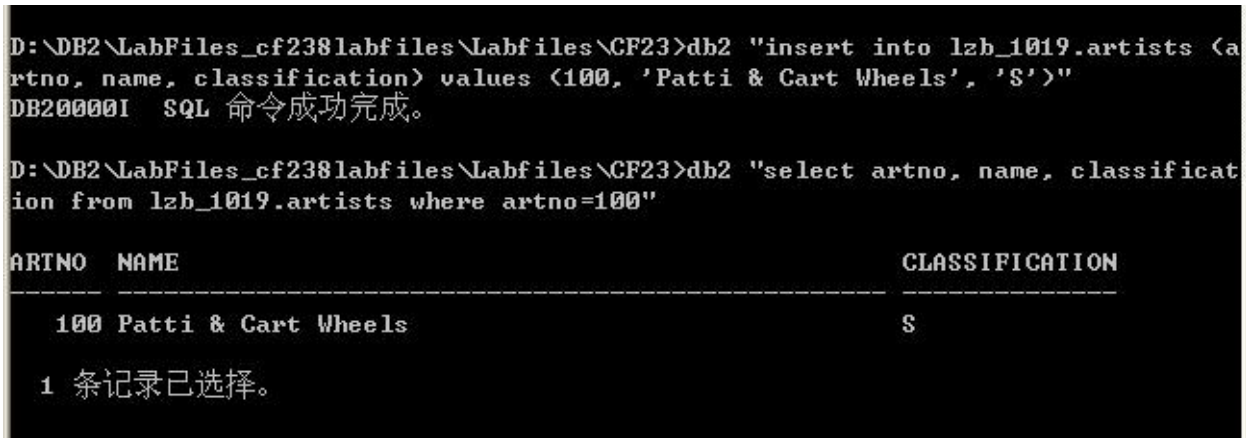
ITEMNO TS                          3
------ --------------------------- ----------------------------------------
   300 2026-10-11-15.34.02.305000 00001D00023LZB_1019.STOCK.FKITEMNO
   300 2026-10-11-15.34.02.305000 00001D00023LZB_1019.STOCK.FKITEMNO

  2 条记录已选择。
```

11. The rows that were moved from the STOCK table to STOEXP and from the ALBUMS table to ALBEXP were foreign key rows that did not have a matching parent key row in the ARTISTS table. You wish to carry this artist's work. Execute SQL to insert a row for item number 100, title of “Patti & Cart Wheels”,

classification of “S” into the ARTISTS table.

Show your work below by printing screen.



12. Now you wish to use the exception table rows in STOEXP and in ALBEXP to insert rows into the STOCK and ALBUMS tables, respectively. View the rows that are in the two exception tables by using the script called selexp. Note that you do not want to insert the N+1 and N+2 columns into the STOCK and ALBUMS tables. Examine the MSG column in the two exception tables.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select title, artno, itemno, ts
, substr(msg,1,40) as msg from lzb_1019.albexp"

TITLE                                              ARTNO  ITEMNO TS
                         MSG
-------------------------------------------------- ------ ------ --------------
-------------- ----------------------------------------
Hand Stands                                           100    300 2026-10-11-15.3
4.02.305000 00001F00023LZB_1019.ALBUMS.FKARTNO
Flips                                                 100    301 2026-10-11-15.3
4.02.305000 00001F00023LZB_1019.ALBUMS.FKARTNO

  2 条记录已选择。


D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select itemno, type, price, qty
, ts, substr(msg,1,40) as msg from lzb_1019.stoexp"

ITEMNO TYPE PRICE   QTY         TS                         MSG

------ ---- ------- ----------- -------------------------- ----------------------
--------------------
   300 C       0.99          99 2026-10-11-15.34.02.305000 00001D00023LZB_1019.S
TOCK.FKITEMNO
   300 C       0.99          99 2026-10-11-15.34.02.305000 00001D00023LZB_1019.S
TOCK.FKITEMNO

  2 条记录已选择。
```

13. Run a script called insexp to insert the rows in the exception tables into the STOCK and ALBUMS tables. The script also selects from the ALBUMS and STOCK tables to see if the rows were inserted properly.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 -tvf INSEXP
connect to lzb_1019

   数据库连接信息

 数据库服务器          = DB2/NT 9.7.0
 SQL 授权标识          = LZB_1019
 本地数据库别名        = LZB_1019


insert into albums select title, artno, itemno from albexp
DB20000I  SQL 命令成功完成。

insert into stock select itemno, type, price, qty from stoexp
DB20000I  SQL 命令成功完成。

select * from albums where artno = 100

TITLE                                              ARTNO  ITEMNO
-------------------------------------------------- ------ ------
Hand Stands                                           100    300
Flips                                                 100    301

  2 条记录已选择。


select * from stock where itemno = 300

ITEMNO TYPE PRICE   QTY
------ ---- ------- -----------
   300 C       0.99          99
   300 C       0.99          99

  2 条记录已选择。
```

Section 7 - Understanding CHECK CONSTRAINT Enforcement

1. Previously you created a check constraint on the TYPE column in the STOCK

table. Review the constraint requirements.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select constname, tabname, subs
tr(text,1,100) as check_condition from syscat.checks where tabschema=user and co
nstname='CCTYPE'"

CONSTNAME
                                                TABNAME

                  CHECK_CONDITION

------------------------------------------------------------------------------
------------------------------------------------ ------------------------------
------------------------------------------------------------------------------
---------------- ------------------------------------------------------------
----------------------------------------

CCTYPE
                                                STOCK

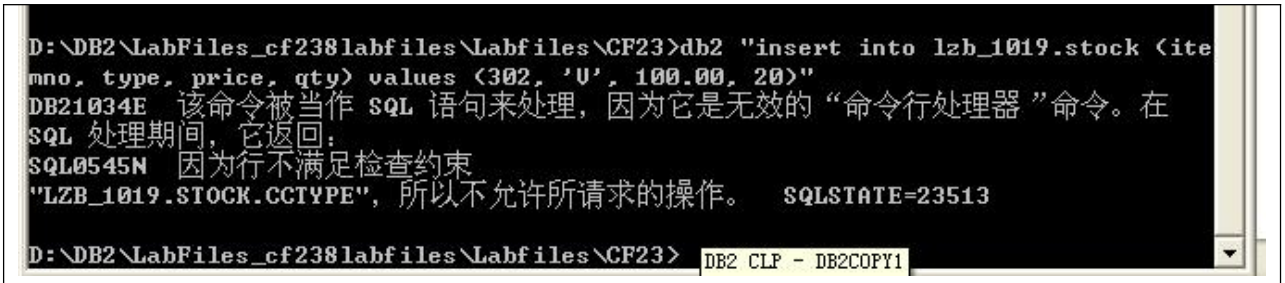
                  TYPE in ('D', 'C', 'R')


  1 条记录已选择。
```

2. The hot new band, Double Dare, has made a Music Video that they want you to carry. Connect to the database and attempt to insert stock data for this artist with an ITEMNO of 302, TYPE of V, PRICE of 100.00, and QTY of 20.

Why did you get the message SQL0545N?

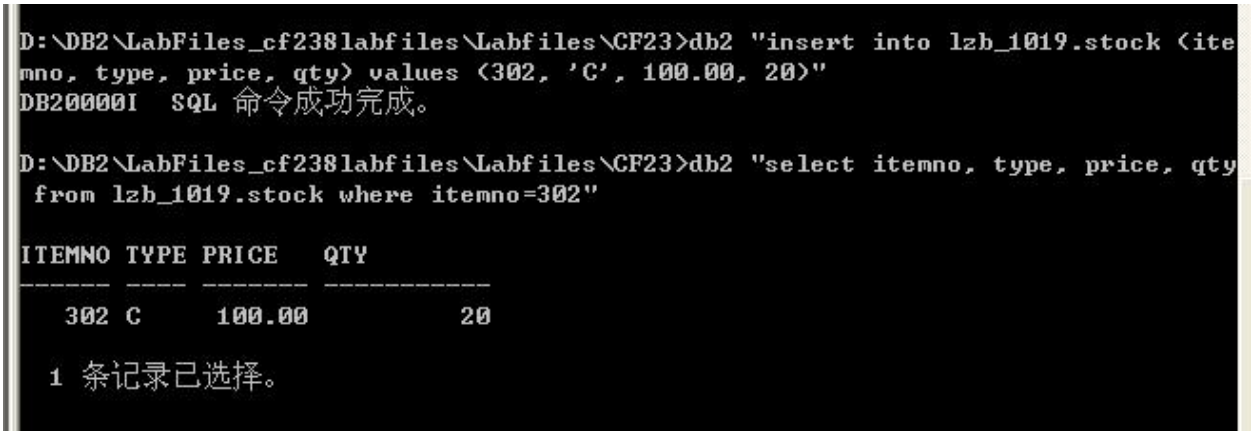
Show your work below by printing screen.



之所以收到错误信息 SQL0545N，是因为试图插入的数据中 `TYPE = 'V'`（录像带 Video）。而在 `STOCK` 表定义的 `CCTYPE` 检查约束条件严格限制为 `TYPE in ('D', 'C', 'R')`。值 `'V'` 不在允许的枚举值集合范围内，违反了该表上的表级检查约束，因此 DB2 约束检查引擎强制拒绝了此插入操作并回滚。

3. Reissue the insert SQL statement with the TYPE = 'C'. You have decided to only carry the CD type.

Show your work below by printing screen.



Section 8 - Understanding TRIGGER Enforcement

1. Previously you created a trigger on the QTY column in the STOCK table. Review the trigger requirements, by selecting from the SYSCAT.TRIGGERS view. Under what conditions would the trigger be fired?

Answer question and show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select substr(trigname,1,12) as
 trigname, substr(tabname,1,10) as tabname, trigtime, trigevent, granularity, su
bstr(text,1,200) as def from syscat.triggers where trigschema=user and trigname=
'REORDER'"

TRIGNAME     TABNAME    TRIGTIME TRIGEVENT GRANULARITY DEF



------------ ---------- -------- --------- ----------- --------------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
---------------
REORDER      STOCK      A        U         R           create trigger reorder af
ter update of qty on stock referencing new as n for each row mode db2sql when (n
.qty <= 5) insert into reorder values (n.itemno, current timestamp)

  1 条记录已选择。
```

触发器定义：

CREATE TRIGGER REORDER

```
AFTER UPDATE OF QTY ON STOCK

REFERENCING NEW AS N

FOR EACH ROW

MODE DB2SQL

WHEN (N.QTY <= 5)

INSERT INTO REORDER VALUES (N.ITEMNO, CURRENT TIMESTAMP);
```

触发条件（Conditions under which the trigger will fire）：

1. 操作类型（Event）：必须是对 `STOCK` 表执行 `UPDATE` 操作；

2. 受影响列（Column）：UPDATE 必须显式涉及 `QTY`（库存量）列；

3. 触发时机（Time）：在更新成功应用之后（`AFTER`）；

4. 粒度（Granularity）：针对每一行分别求值（`FOR EACH ROW`）；

5. 谓词条件（WHEN 子句）：更新后的新库存值必须满足 `N.QTY <= 5`。注：普通的 INSERT、DELETE、或更新除 QTY 外的其他字段、以及将 QTY 更新为大于 5 的数值均绝对不会触发此触发器。

2. The Double Dare CD, ITEMNO 302, entitled “I Dare You” is selling very fast. Update the STOCK table to reflect the current inventory of 3. Note that no messages occur when a trigger is fired.

Show your work below by printing screen.



3. Query the REORDER table to see if the TRIGGER was fired during the previous update to the QTY in the STOCK table for ITEMNO 302.

Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select itemno, timestamp from l
zb_1019.reorder where itemno = 302"

ITEMNO TIMESTAMP
------ --------------------------
   302 2026-10-11-15.54.21.820000

  1 条记录已选择。
```

查询成功返回了包含 `ITEMNO = 302` 以及当前系统时间戳的一条记录。这确凿证明了当执行 UPDATE 将库存量从 20 修改为 3 时，满足了 `WHEN (N.QTY <= 5)` 触发条件，`REORDER` 触发器被成功激活并自动向 `REORDER` 补货表中执行了 INSERT 操作。

4. Previously you created an alias called EMPTYSTOCK for the REORDER table. Does it give the same results as querying directly against REORDER?
Answer question and show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select * from lzb_1019.reorder"

ITEMNO TIMESTAMP
------ --------------------------
   302 2026-10-11-15.54.21.820000

  1 条记录已选择。

D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 "select * from lzb_1019.emptysto
ck"

ITEMNO TIMESTAMP
------ --------------------------
   302 2026-10-11-15.54.21.820000

  1 条记录已选择。
```

是的，查询别名 `EMPTYSTOCK` 与直接查询 `REORDER` 返回完全一致的结果（Yes, it gives the exact same results）。原理剖析：别名（Alias）是 DB2 中指向底层实际物理表的一个指针或符号名称，它不会创建独立的数据副本，也不占用额外的物理存储空间。当客户端通过别名 `EMPTYSTOCK` 发

起查询时，数据库引擎在解析执行计划阶段会自动将其透明解析重定向到底层引用的基表 `REORDER` 上，因此二者的查询结果与语义完全等价。

Section 9 - Using the db2look tool

1. This is a command line tool you will use to extract the DDL for the ARTISTS table into a file. You would do this to create a look-alike table in another database, or to store the structure away for future use.

Show your work below by printing screen.



```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>type artists.ddl
-- 此 CLP 文件是使用 DB2LOOK 版本 "9.7" 创建的
-- 时间戳记: 2026-10-11 15:56:35
-- 数据库名称: LZB_1019
-- 数据库管理器版本: DB2/NT Version 9.7.0
-- 数据库代码页: 1208
-- 数据库整理顺序为: SYSTEM_1386


CONNECT TO LZB_1019;

------------------------------------------------
-- 表的 DDL 语句 "LZB_1019"."ARTISTS"
------------------------------------------------


CREATE TABLE "LZB_1019"."ARTISTS"  (
                  "ARTNO" SMALLINT NOT NULL ,
                  "NAME" VARCHAR(50) ,
                  "CLASSIFICATION" CHAR(1) NOT NULL ,
                  "BIO" CLOB(102400) LOGGED COMPACT ,
                  "PICTURE" BLOB(512000) NOT LOGGED COMPACT )
                 IN "DMS01" INDEX IN "DMS02" LONG IN "DMS03" ;


-- 表上主键的 DDL 语句 "LZB_1019"."ARTISTS"

ALTER TABLE "LZB_1019"."ARTISTS"
        ADD PRIMARY KEY
                ("ARTNO");


-- 基于表的别名的 DDL 语句 "LZB_1019"."ARTISTS"

CREATE ALIAS "LZB_1019"."SINGERS" FOR TABLE "LZB_1019"."ARTISTS";









----------------------------

-- 视图的 DDL 语句

----------------------------
SET CURRENT SCHEMA = "LZB_1019";
SET CURRENT PATH = "SYSIBM","SYSFUN","SYSPROC","SYSIBMADM","LZB_1019";
create view music as select albums.title, artists.classification, artists.name
from albums, artists where artists.artno = albums.artno;
```

```
COMMIT WORK;

CONNECT RESET;

TERMINATE;
```

2. Use the DB2 DESCRIBE tool and compare the results with the DDL file.
Show your work below by printing screen.

```
D:\DB2\LabFiles_cf238labfiles\Labfiles\CF23>db2 describe table lzb_1019.artists
show detail



                                                                        列
                列          分区键              代码
列名
                                                    数据类型模式

                            数据类型名称
 长          小数位     NULL     号          序列                  页        缺省值
------------------------------------------------------------------------------
------------------------------------------------ --------------------------------
------------------------------------------------------------------------------
---------------- ---------------------------------------------------------------
------------------------------------------------------------------ ----------- -
------- -------- ---------- ------------------ -------- ----------------------
-----------------
ARTNO
                                                SYSIBM

                SMALLINT
                                                                              2
      0 否              0                  0        0


NAME
                                                SYSIBM

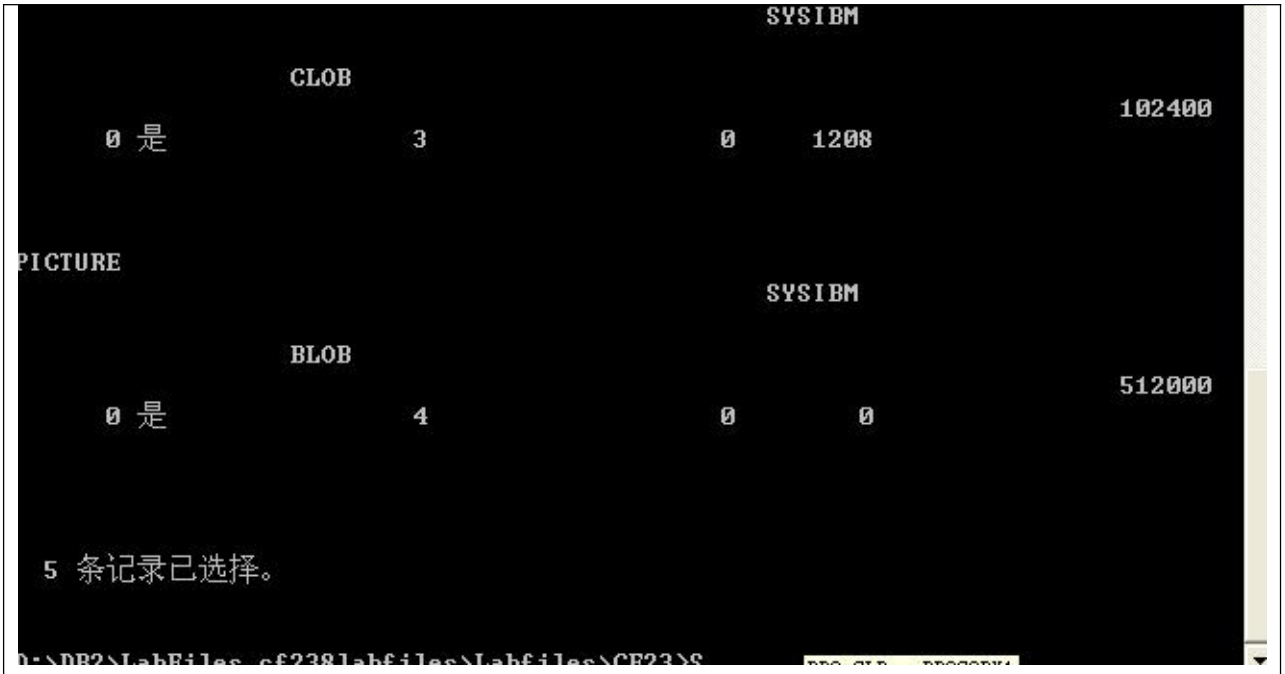
                VARCHAR
                                                                             50
      0 是              1                  0     1208


CLASSIFICATION
                                                SYSIBM

                CHARACTER
                                                                              1
      0 否              2                  0     1208


BIO
                                                SYSIBM

                CLOB
```



结构比对维度	db2 describe table 结果	db2look 导出的 DDL 结果	比对结论
列定义数量	共 5 个列（ARTNO, NAME, CLASSIFICATION, BIO, PICTURE）	CREATE TABLE 中列举的 5 个列相同	完全吻合
主键约束	ARTNO 标记为 NULL = 否	包含 ADD CONSTRAINT CCARTNO PRIMARY KEY (ARTNO)	完全吻合
LOB 大对象属性	BIO 为 CLOB(102400)，PICTURE 为 BLOB(512000)	包含对应字节大小，并声明了 LOGGED COMPACT / NOT LOGGED 属性	完全吻合
物理存储映射	表元数据中指向三个表空间	显式输出 IN DMS01 INDEX IN DMS02 LONG IN DMS03	完全吻合

db2 describe 以表格形式清晰列出了各列的数据类型、物理长度以及可空性；而 db2look 则更进一步，逆向重构出了完整的 SQL DDL 脚本，不仅包含了字段定义，还还原了主键约束名、表空间分配子句（数据在 DMS01、索引在 DMS02、长对象在 DMS03）以及 LOB 存储参数。两者的底层元数据完全一致，互为印证。